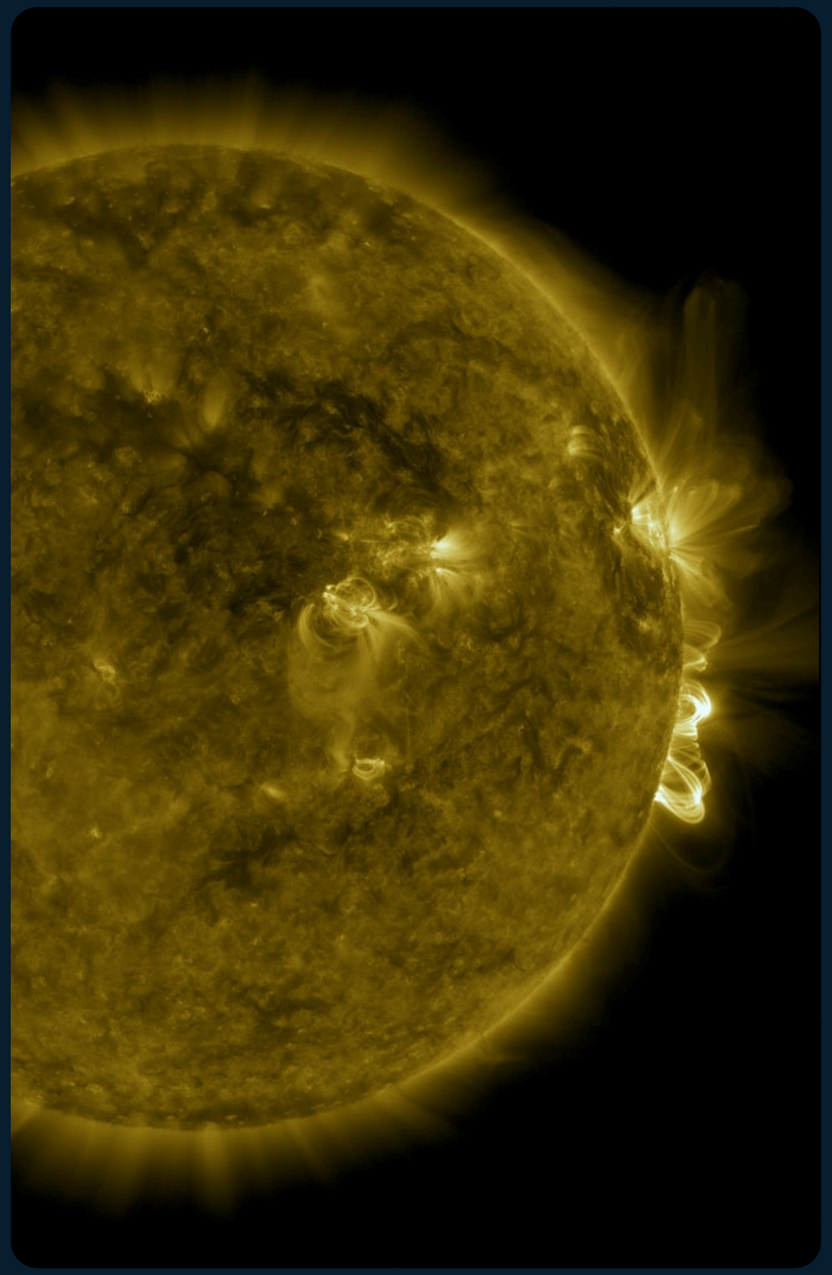
Solar Astrophysics School · October 2026

HelioData Academy 2026

Stochastic Processes in Solar Flares

From collisional noise to hard X-ray spectra

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Session roadmap

WHERE WE ARE GOING

01

What we observe

Thermal and non-thermal X-rays, the Neupert effect, and why fluids are not enough.

02

The stochastic language

Collisions as random kicks: Langevin for one electron, Fokker-Planck for the population.

03

Acceleration and X-rays

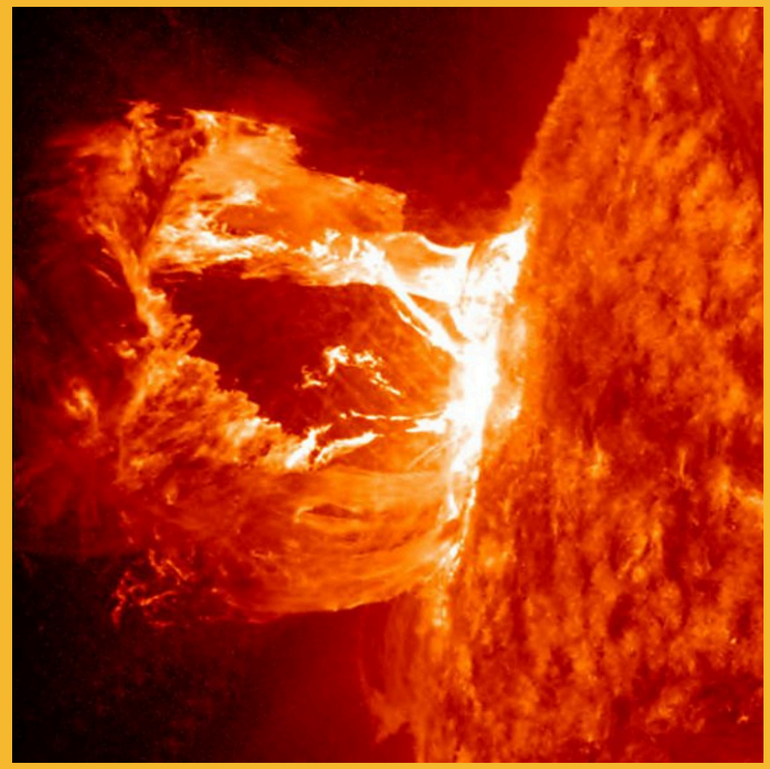
How power-law tails form, and how bremsstrahlung lets us read them.

SECTION

01

What we observe

Every flare shows two electron populations.
Where do they come from?



Flares are particle accelerators

- Up to 10^{32} erg released in minutes
- Driven by magnetic reconnection
- Much of it goes into electrons of 10 -100 keV
- They shine in hard X-rays

Hard X-rays are our window into the accelerated electrons!

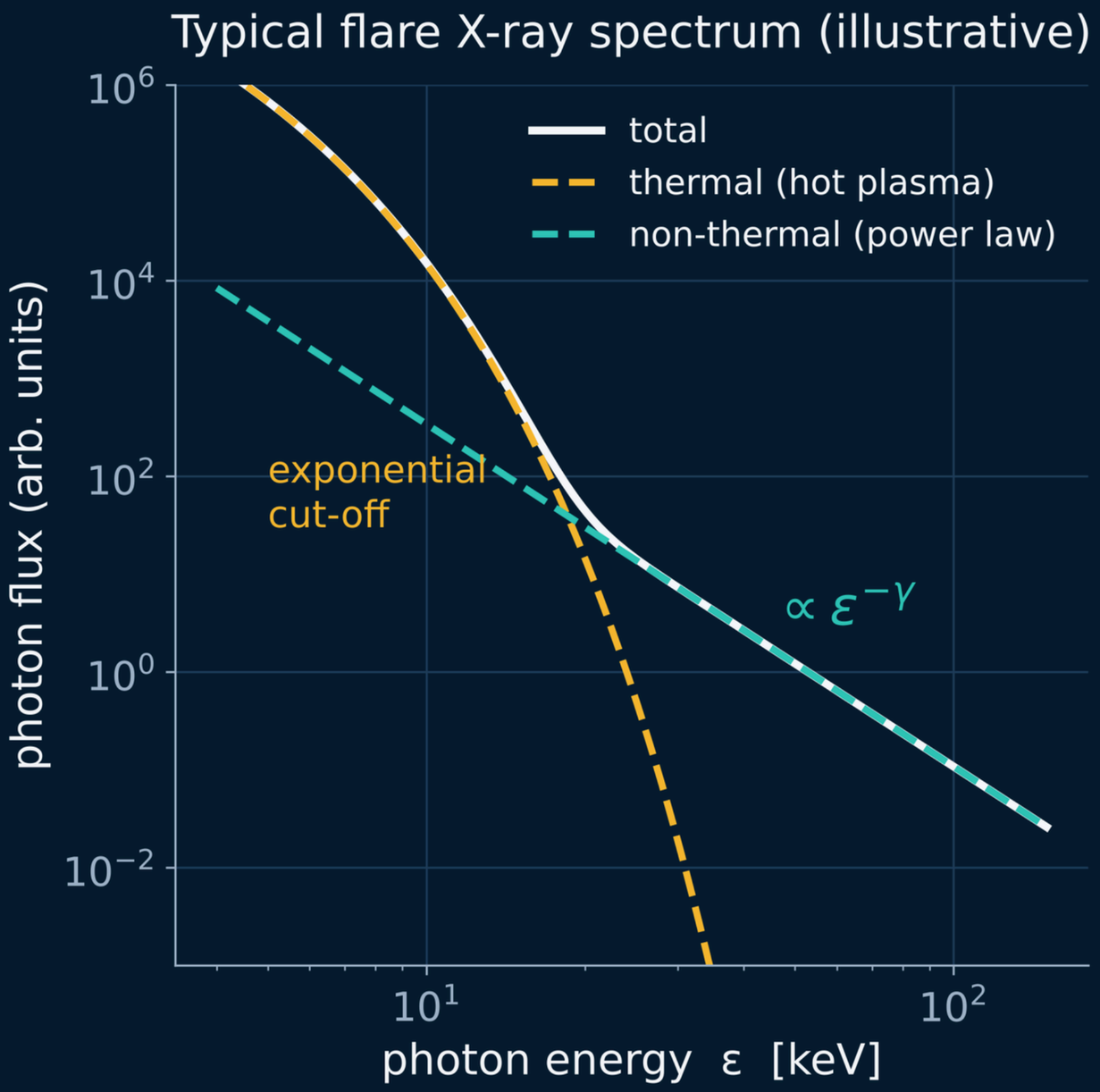


Figure: own work. Illustrative model (thermal + power law)

Two kinds of X-ray emission

Thermal vs. non-thermal:

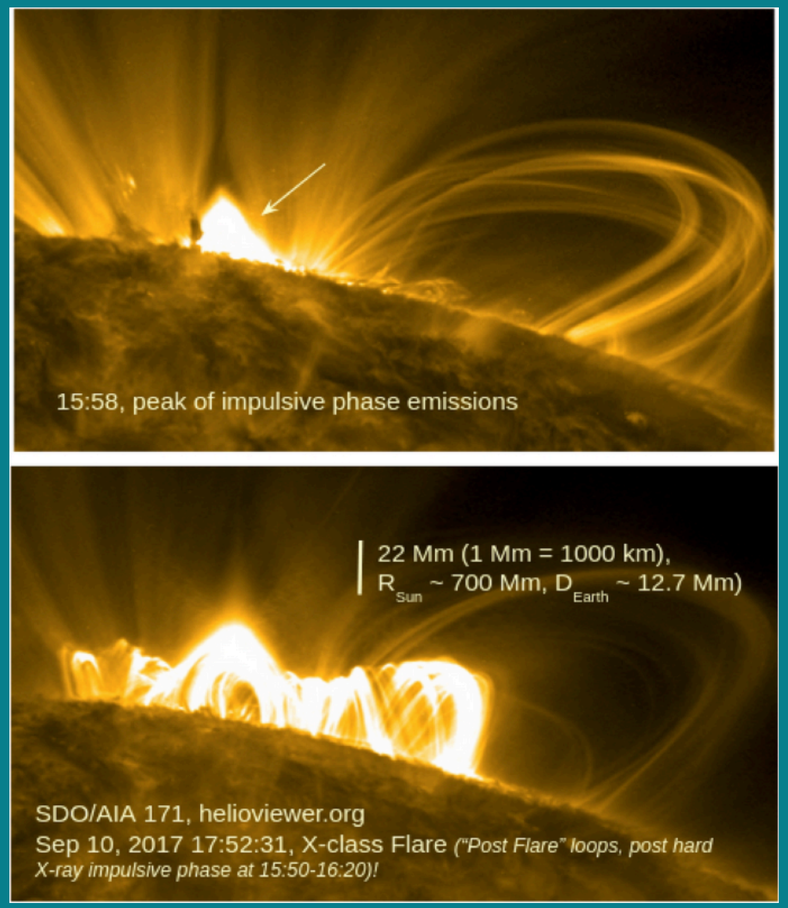
Thermal

- Dominates below ~15 - 20 keV
- Gradual, smooth time profile
- Exponential spectrum: hot plasma at 10 – 30 MK

Non-thermal

- Dominates above ~ 20 keV
- Impulsive spikes
- Power-law spectrum ε - γ: accelerated electrons

Where do these two populations come from?



$$\frac{dF_{\text{thermal}}}{dt}(t) \propto F_{\text{non-thermal}}(t)$$

The Neupert effect

The time derivative of the thermal (soft X-ray) flux follows the non-thermal (hard X-ray) flux. Electrons heat the chromosphere, it evaporates and fills the loops with hot plasma: the two populations are causally linked.

Reference: Neupert (1968), ApJ 153, L59

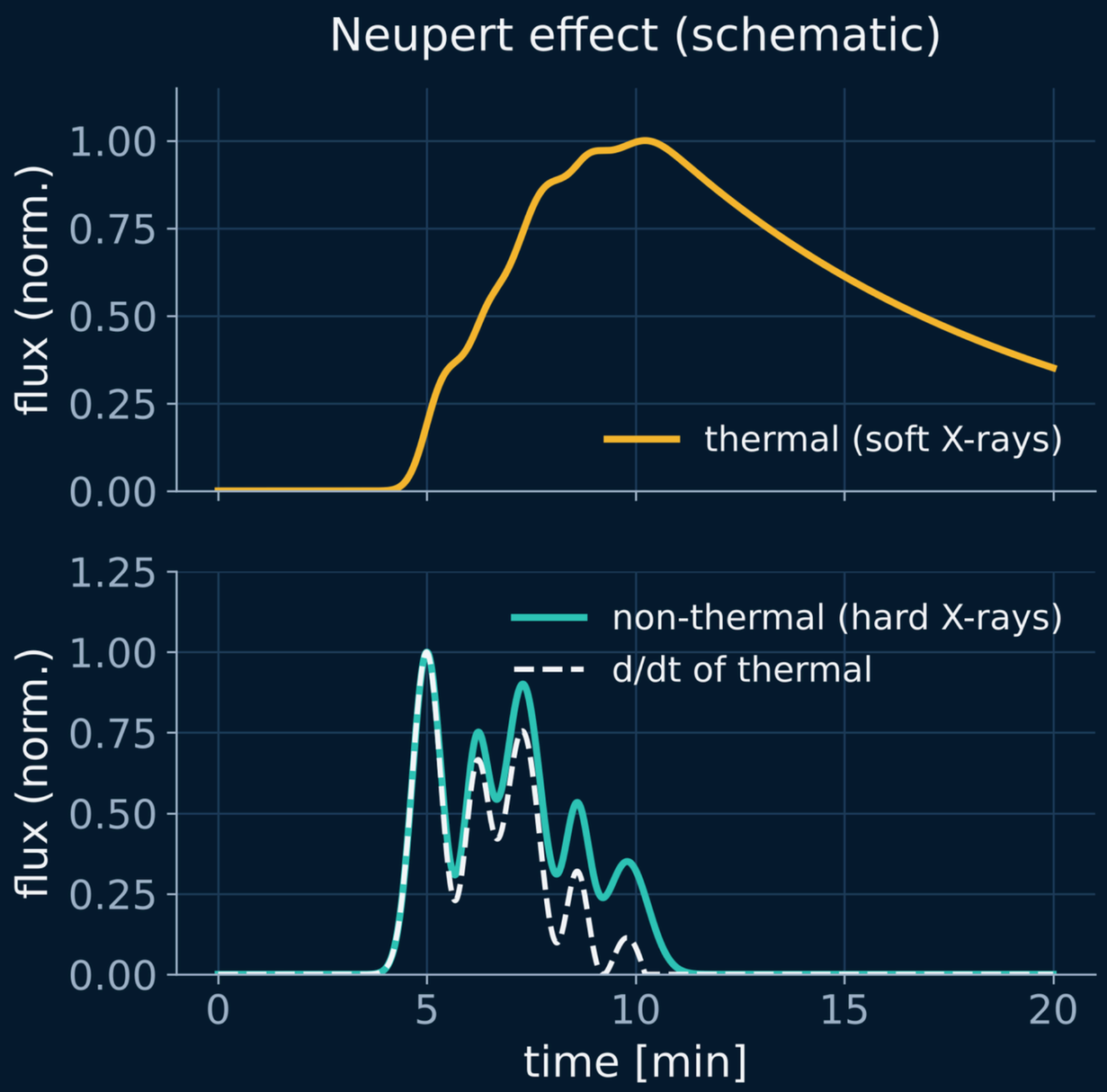


Figure: own work. Schematic light curves

Why fluids are not enough?

01

20 keV

Typical energy of an electron that emits hard X-rays

02

8×10^{19} cm–2

Column it crosses before stopping: $Ns = E^2/2K$

03

10^{10} cm–3

Typical density of the flaring corona

04

≈ 80 Mm

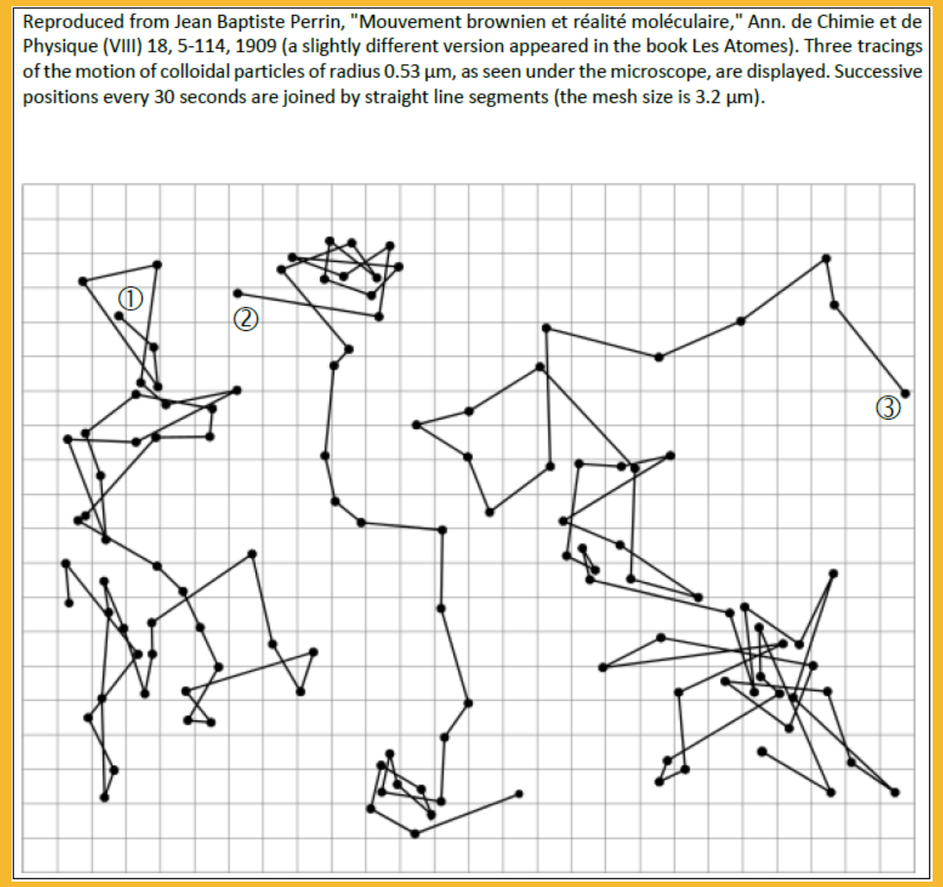
Stopping length Ns/n: longer than a loop. We need f(v), not just n, T

SECTION

02

The stochastic language

Each electron is unpredictable. The population is not.



Collisions are a random process

- The Coulomb force is long-range: each electron feels many particles at once
- Its path is a sum of many small, random deflections
- Small-angle kicks dominate
- Unpredictable one by one, well defined statistically

> Many small random kicks: a stochastic process described by a mean and a variance.

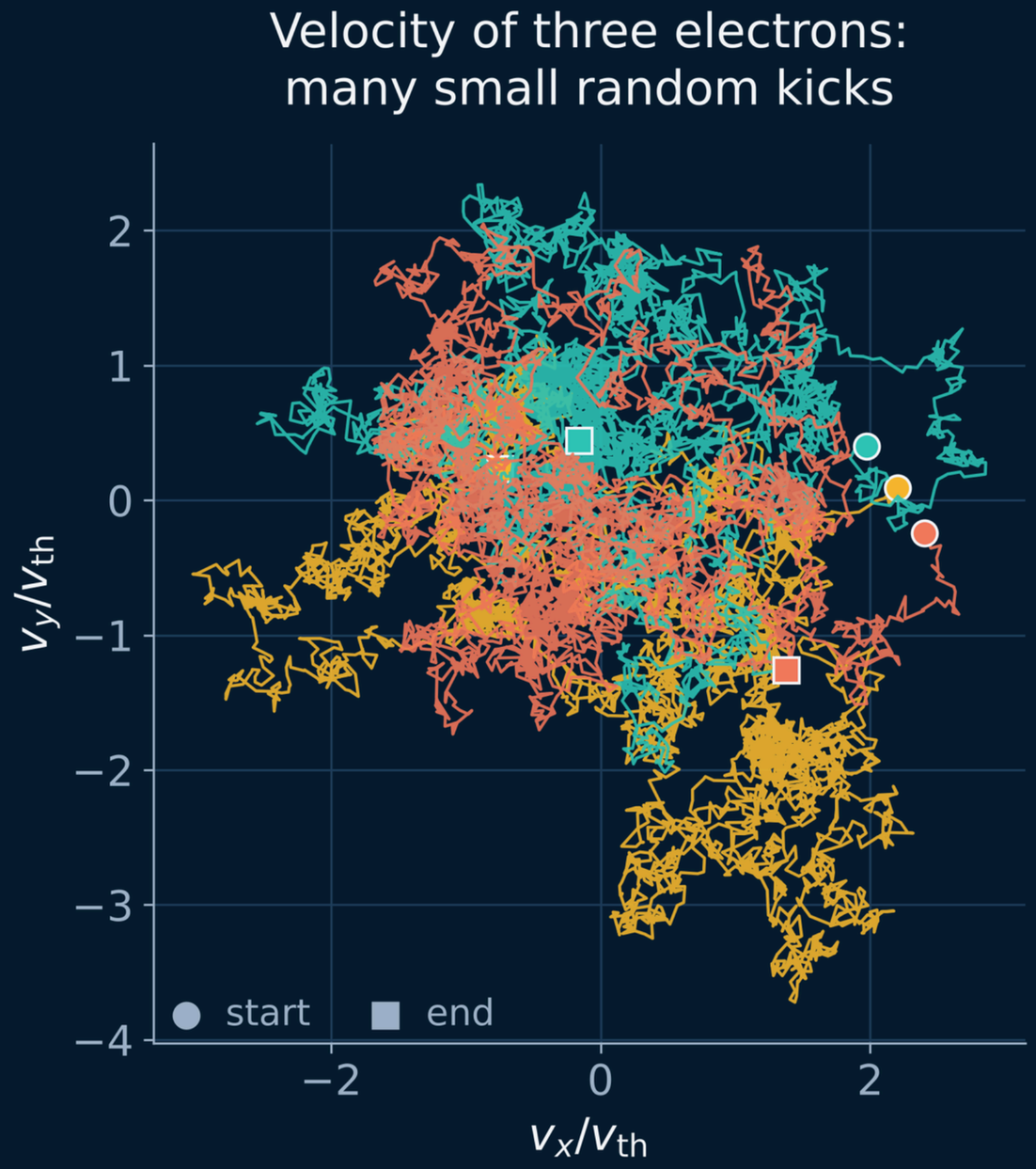


Figure: own work. Illustrative model.

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x}\left[\mu(x)\,P\right] + \frac{\partial^2}{\partial x^2}\left[D(x)\,P\right]$$

$$\mu = \frac{\langle \Delta x \rangle}{\Delta t} \qquad D = \frac{1}{2}\,\frac{\langle (\Delta x)^2 \rangle}{\Delta t}$$

Fokker–Planck equation (the population)

For a Markov process with small jumps, expand in the jump size (Kramers-Moyal) and keep two terms. Drift μ: mean change per unit time. Diffusion D: half the variance per unit time.

Reference: Risken (1996); Gardiner (2009)

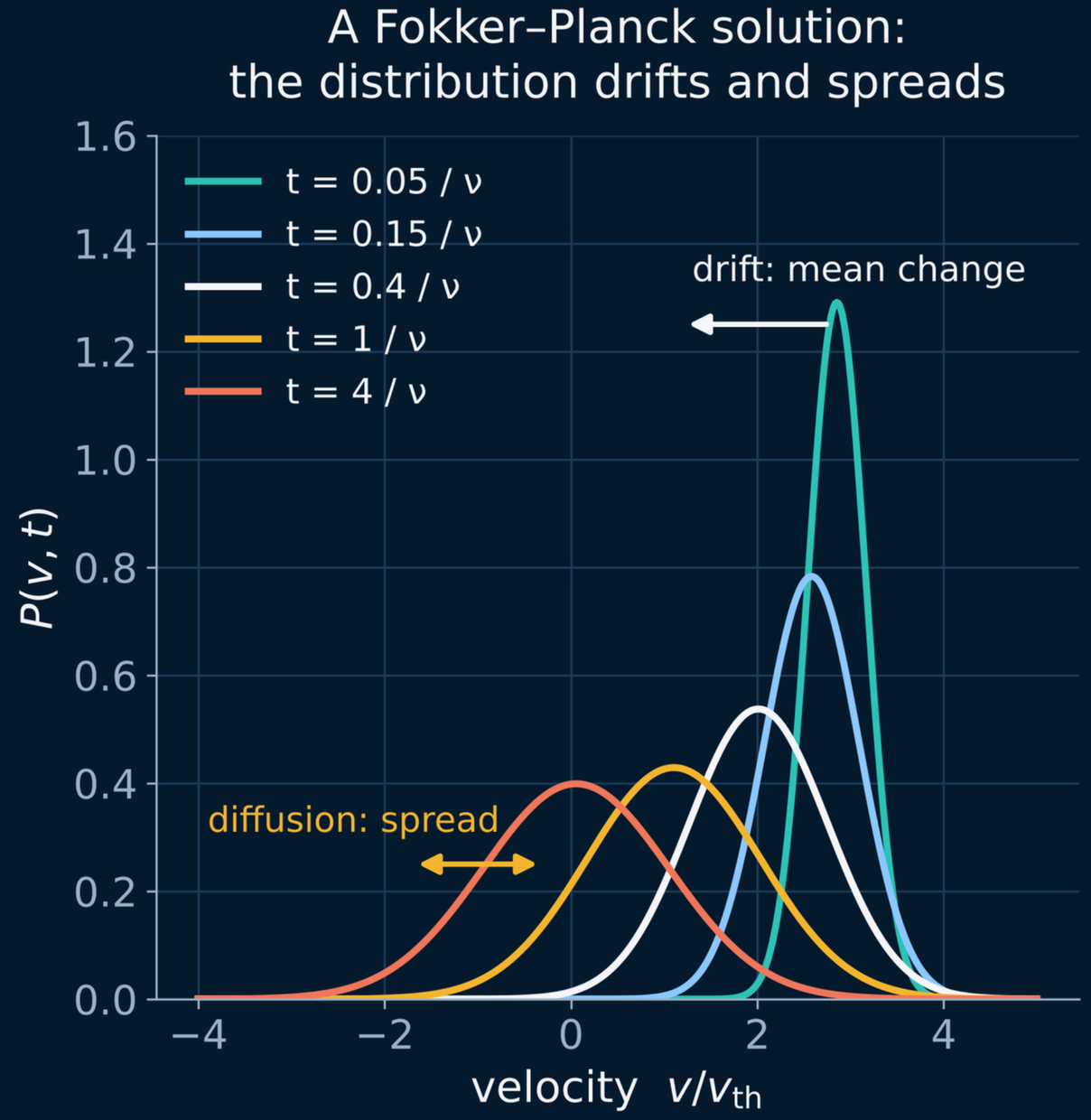


Figure: own work. Illustrative model

$$dx = \mu(x)\,dt + \sigma(x)\,dW_t$$

$$x_{n+1} = x_n + \mu(x_n)\,\Delta t + \sigma(x_n)\sqrt{\Delta t}\,\xi_n$$

Langevin equation (one particle)

Deterministic drag plus random kicks. dWt is Gaussian with mean 0 and variance dt. On a computer (Euler-Maruyama): one Gaussian random number per particle per step.

Reference: Kloeden & Platen (1992)

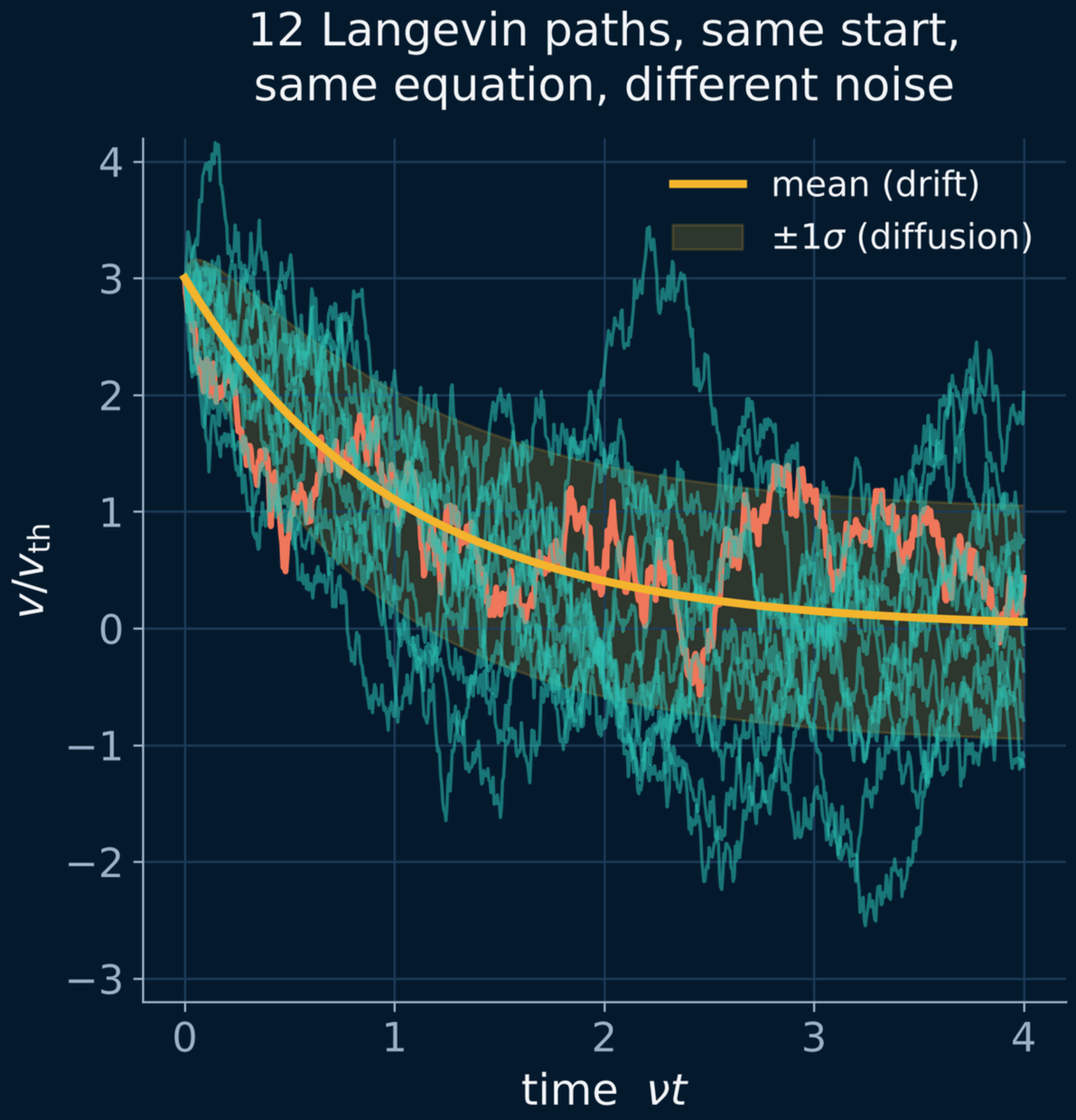


Figure: own work. Illustrative model

Two descriptions, one process

- **Langevin:** follows one electron
- **Fokker-Planck:** follows the population
- Linked by Itô's lemma: $D = \sigma^2/2$
- If D depends on x, an extra drift $D'(x)$ appears (noise-induced drift)

KEY IDEA

The histogram of many Langevin paths is the Fokker-Planck solution!

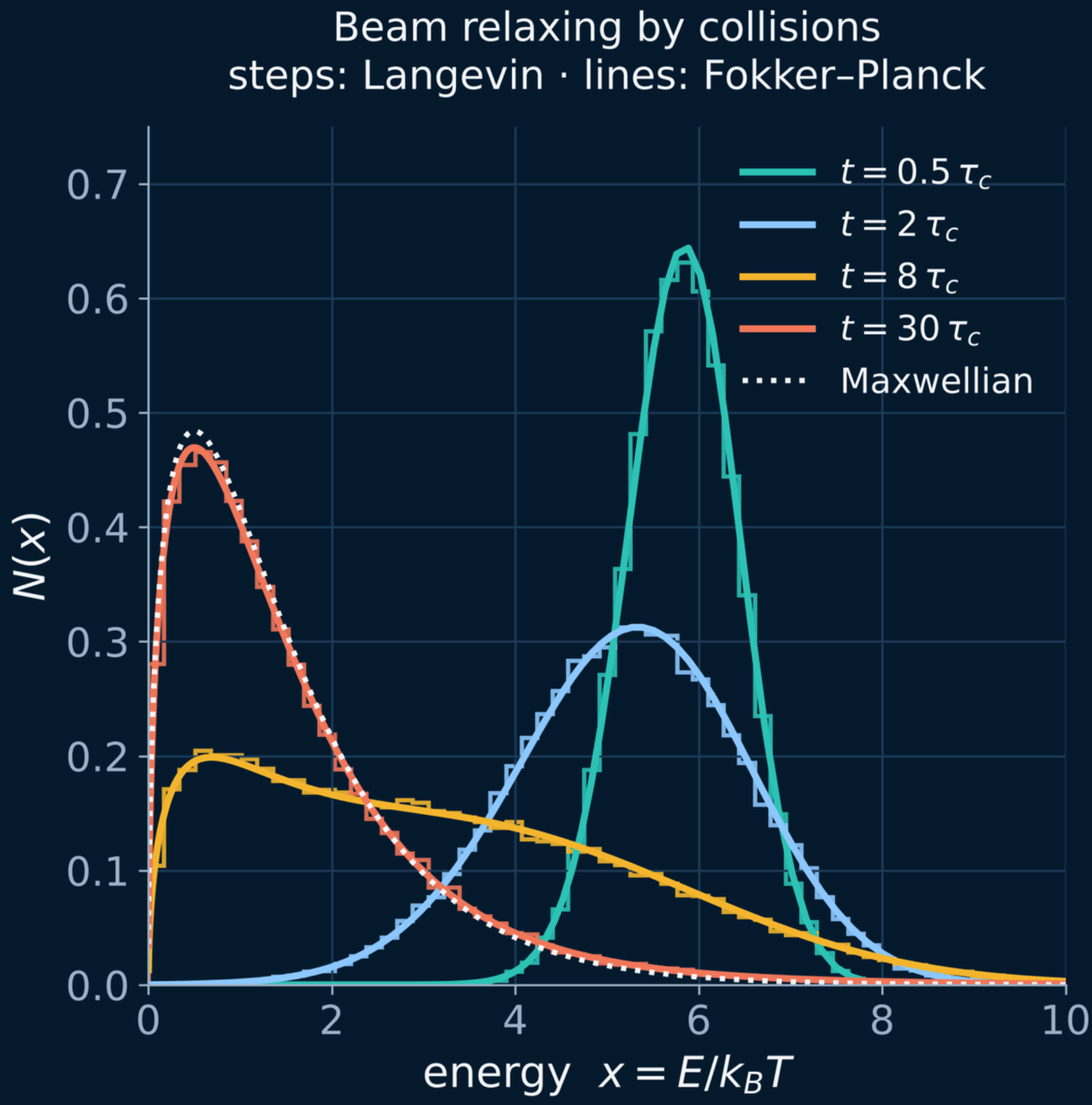


Figure: own work. Illustrative model

$$dv = -\nu v\, dt + \sqrt{2D_v}\, dW_t$$

$$D_v = \nu \frac{k_B T}{m} \quad \Longrightarrow \quad f_{\mathrm{eq}} \propto e^{-mv^2/2k_B T}$$

Fluctuation–dissipation (Einstein relation)

In the Ornstein-Uhlenbeck process the steady state is Gaussian with variance Dv/v. It is the Maxwellian only if drag and kicks are tied by the bath temperature. That is why collisions always lead to thermal equilibrium.

Reference: Uhlenbeck & Ornstein (1930)

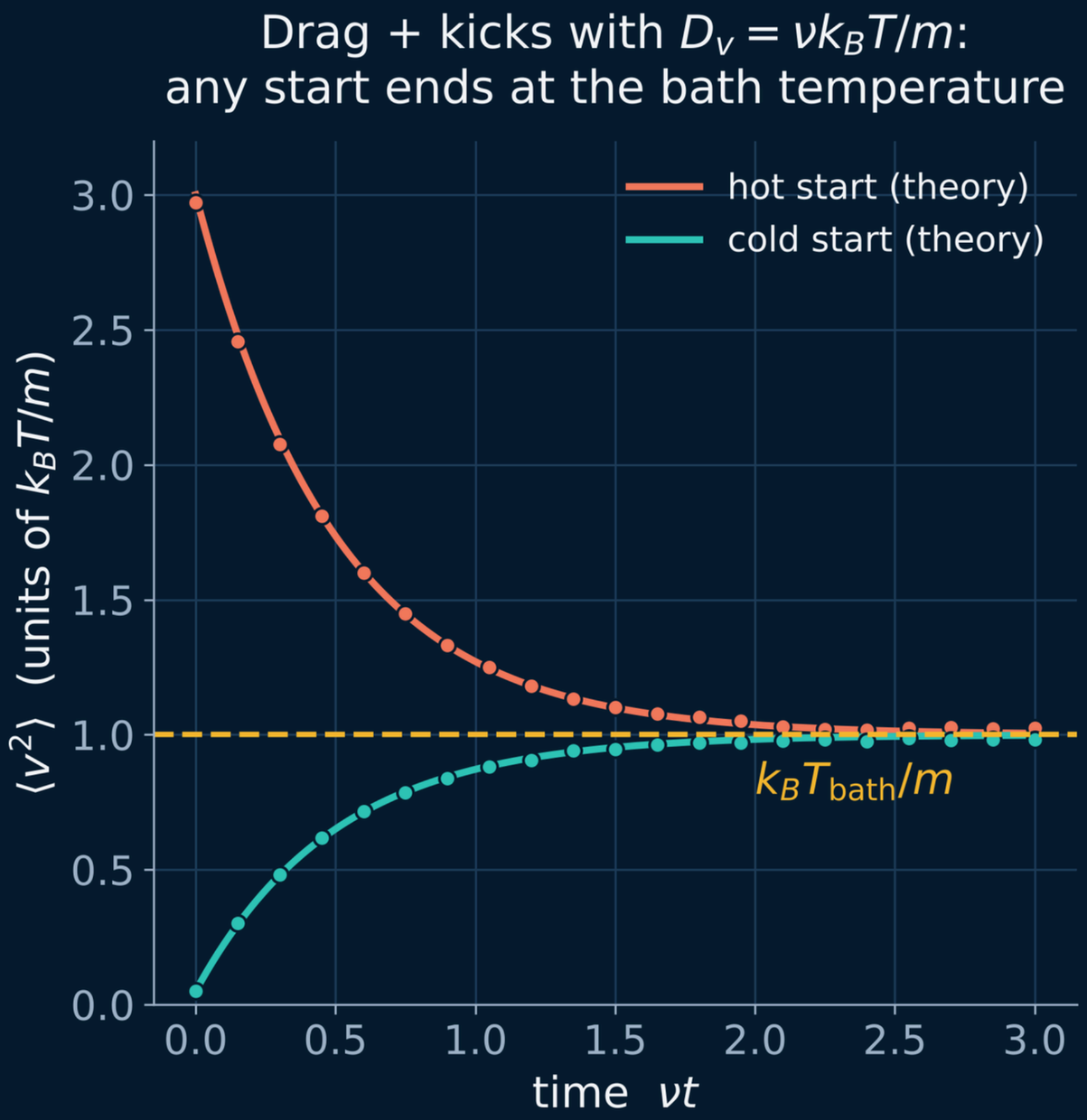


Figure: own work. Illustrative model

$$\nu(v) \;\propto\; \frac{n \ln \Lambda}{v^3}$$

$$\frac{dE}{dN} = -\frac{K}{E}, \qquad K = 2\pi e^4 \ln \Lambda$$

Coulomb collisions weaken with energy

The Rutherford cross-section falls as v-4, so ν ∝ v-3: a fast electron collides less and less as it gains energy. If something feeds it energy faster than collisions remove it, it runs away from the thermal bath.

Reference: Dreicer (1959); Emslie (1978)

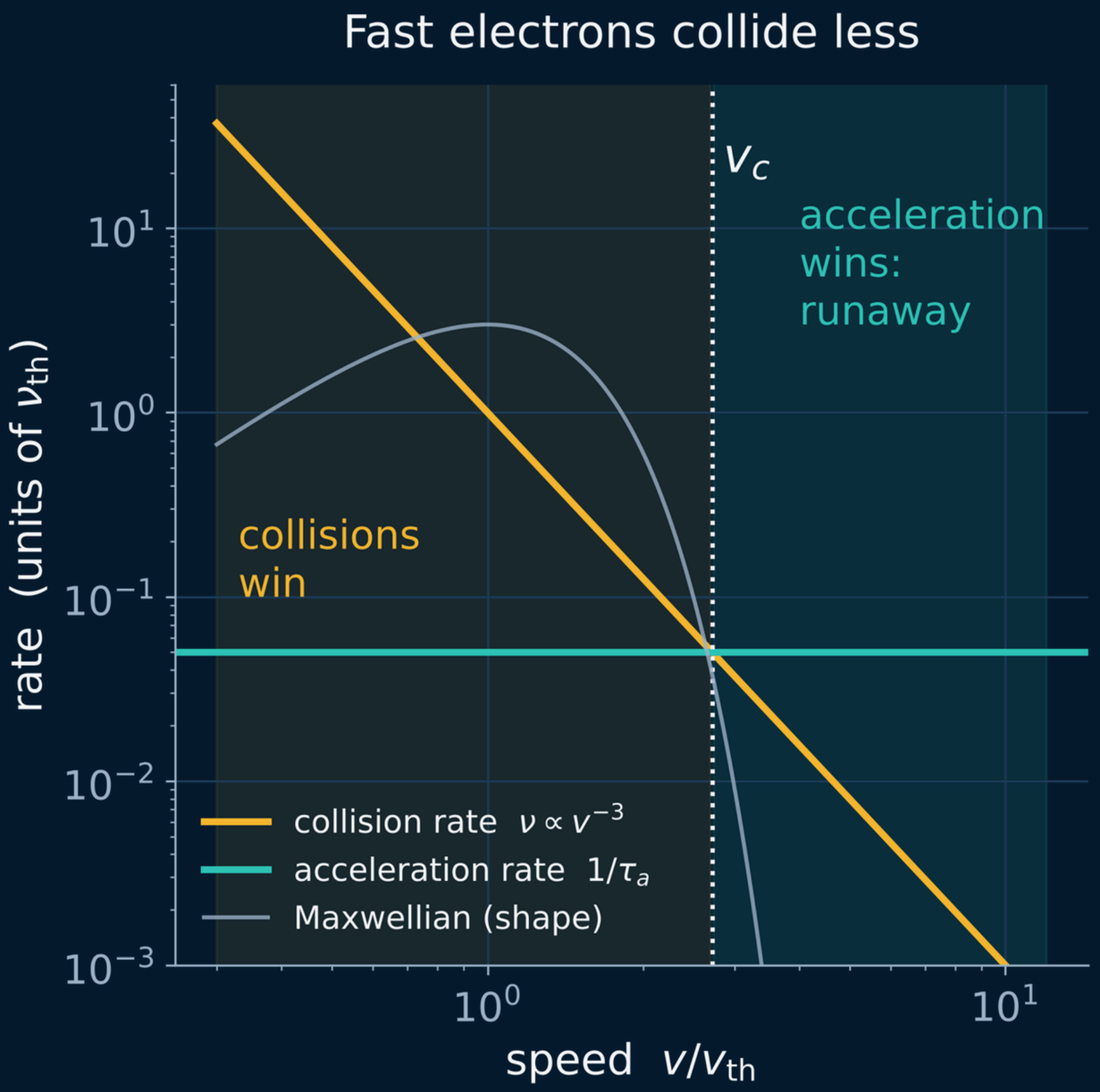


Figure: own work. Illustrative model

Switch on acceleration

- Add stochastic acceleration: $D_a = x^2/\tau_a$, with $x = E/kBT$
- Collisional loss rate ~ x^(-3/2) falls; acceleration rate ~ 1/τa does not
- They cross at $x_c \sim \tau_a^{(2/3)}$
- Below x_c: Maxwellian core. Above x_c: non-thermal tail

Thermal and non-thermal both emerge from the same Fokker-Planck equation.

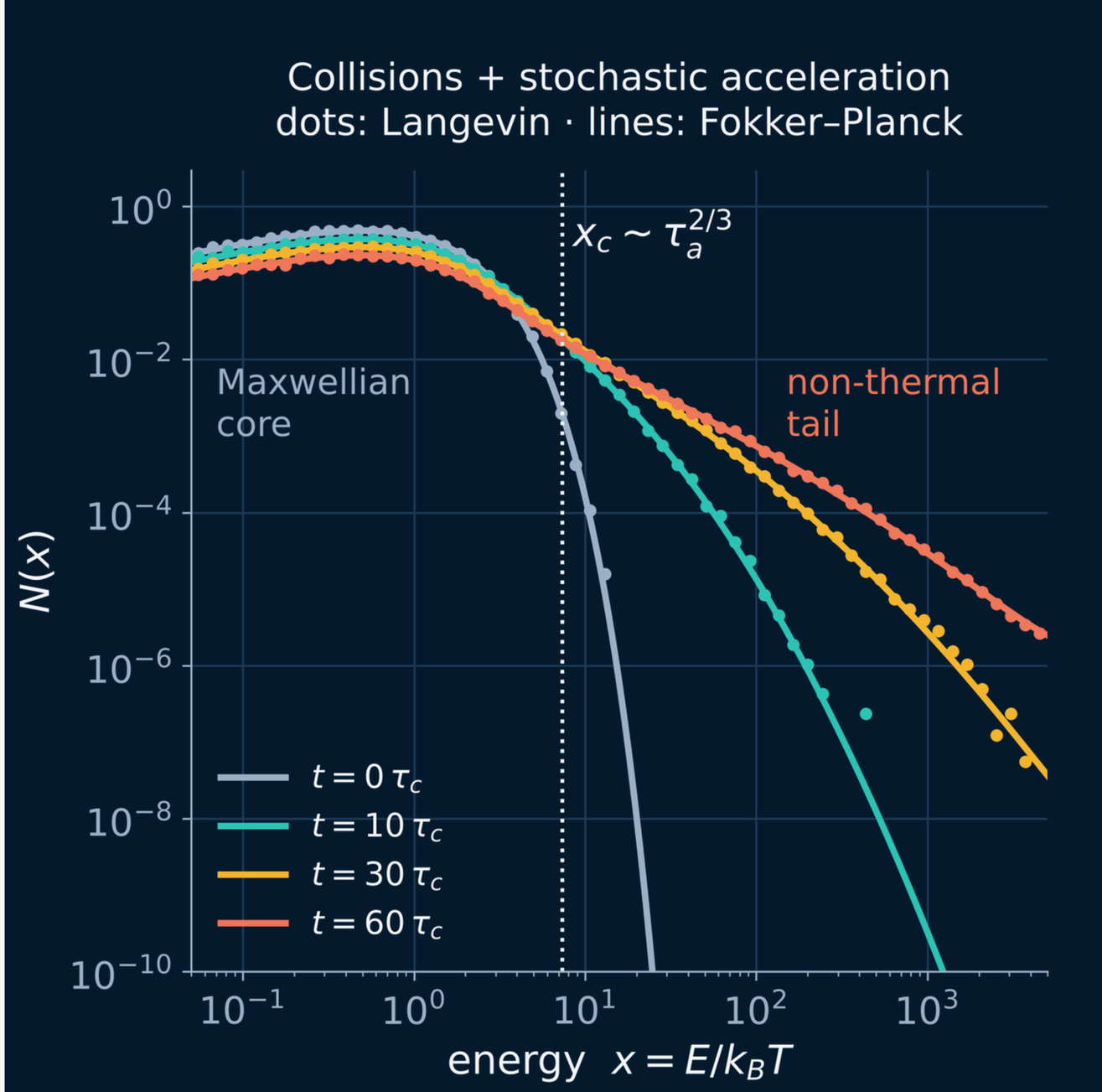


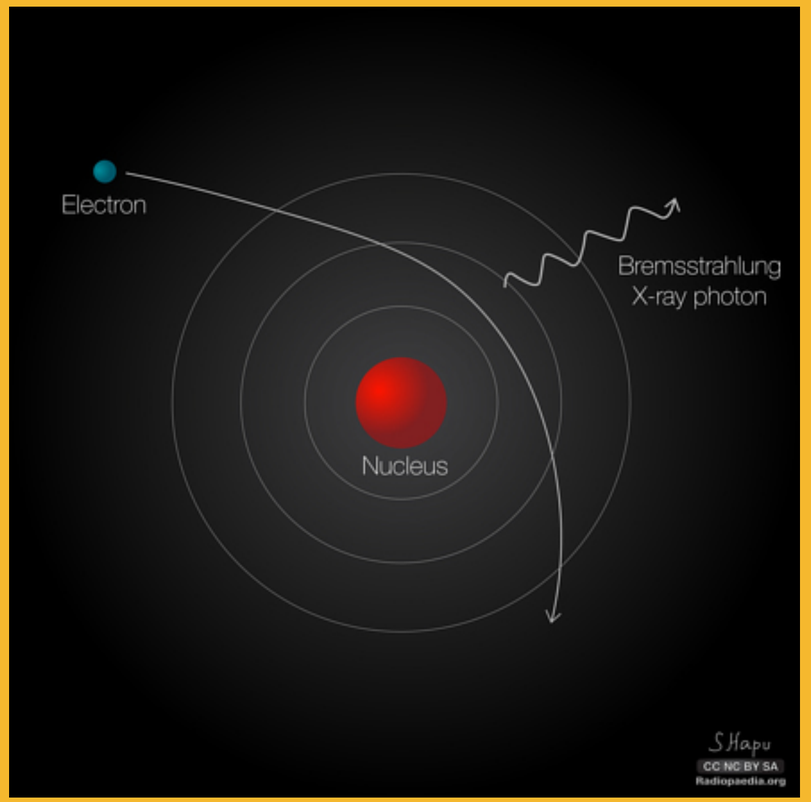
Figure: own work. Illustrative model

SECTION

03

Acceleration and X-rays

How do power laws form, and how do we read them from photons?



How are electrons accelerated?

01

DC electric fields

Reconnection current sheets; Dreicer field → a drift term

02

Shocks · Fermi I

Repeated shock crossings → advection + spatial diffusion

03

Turbulence · Fermi II

Resonant wave-particle scattering → diffusion in momentum

04

One equation

All three are terms of the same Fokker-Planck equation

$$\frac{\partial f}{\partial t} = \frac{1}{p^2}\frac{\partial}{\partial p}\left(p^2 D_{pp}\frac{\partial f}{\partial p}\right) - \frac{f}{\tau_{\rm esc}} + Q$$

$$D_{pp} \sim \frac{p^2}{\tau_a}$$

Stochastic (second-order Fermi) acceleration

Electrons scatter off turbulent waves and gain or lose energy at random, with a net gain at second order. D_pp comes from the turbulence spectrum (quasi-linear theory). It is literally a Fokker-Planck equation.

Reference: Petrosian (2012), Space Sci. Rev. 173, 535

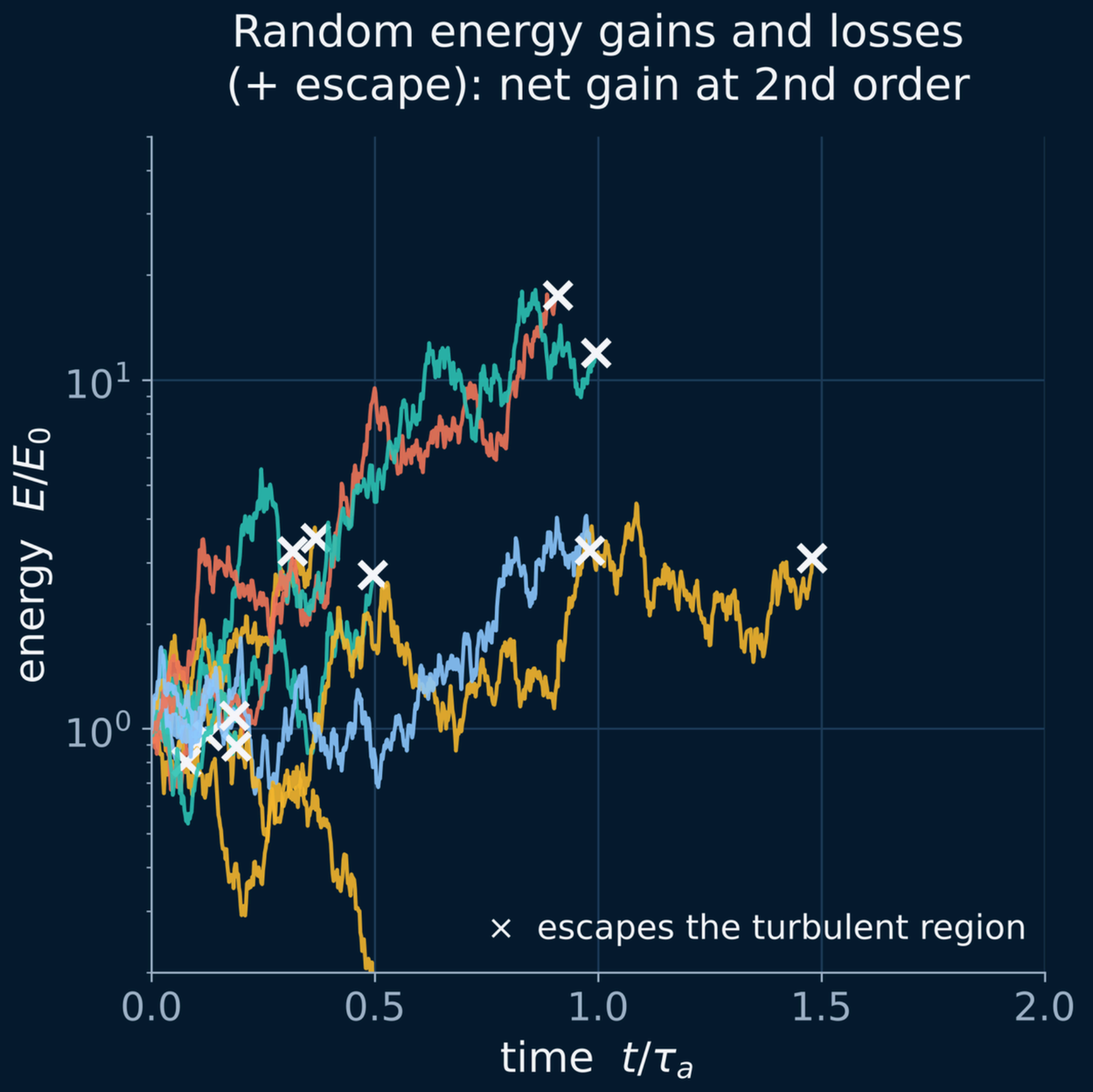


Figure: own work. Illustrative model

$$\frac{\partial N}{\partial t} = \frac{\partial}{\partial E}\left(\frac{E^2}{\tau_a}\frac{\partial N}{\partial E}\right) - \frac{N}{\tau_{\rm esc}} + Q_0\,\delta(E - E_0)$$

$$N \propto E^{-\delta} \implies \boxed{\delta(\delta - 1) = \frac{\tau_a}{\tau_{\rm esc}}}$$

Leaky box: where power laws come from

Inject at E_0, diffuse in energy, escape after τesc. The steady state is a power law whose index is set by a competition of time scales: efficient acceleration gives hard spectra (small δ), fast escape gives soft ones.

Reference: Petrosian (2012); Chen & Petrosian (2013)

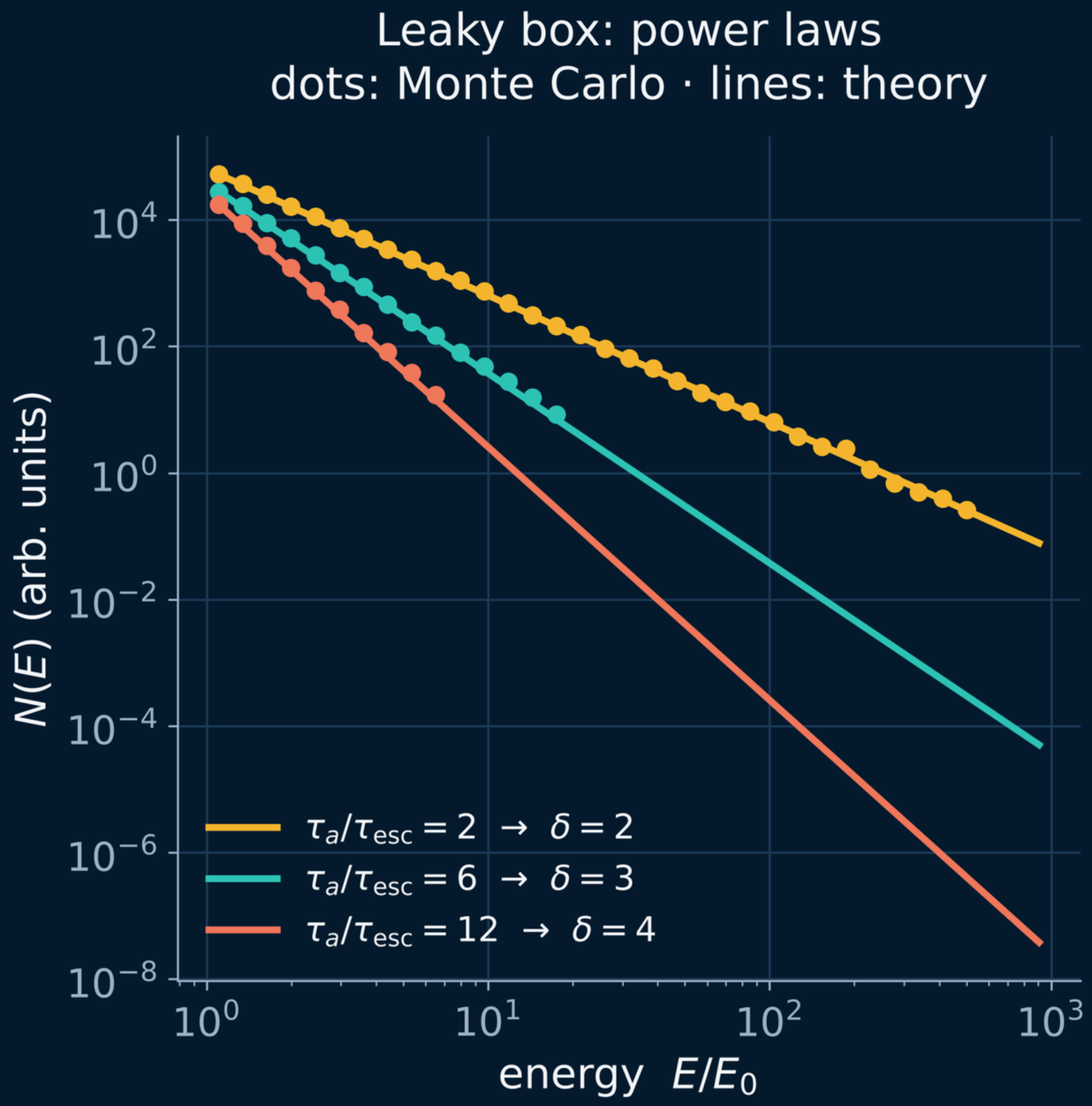


Figure: own work. Illustrative model

$$\sigma(\varepsilon, E) = \frac{\sigma_0}{\varepsilon\, E} \quad (E > \varepsilon)$$

$$I_{\rm th}(\varepsilon) \propto \frac{EM}{\sqrt{T}} \frac{e^{-\varepsilon/k_B T}}{\varepsilon}$$

Bremsstrahlung: from electrons to photons

Electrons deflected by ions emit X-rays; Kramers' cross-section is the simplest model (photon energy ε, electron energy E). A Maxwellian gives an exponential spectrum that curves down in log-log.

Reference: Kramers (1923); Koch & Motz (1959)

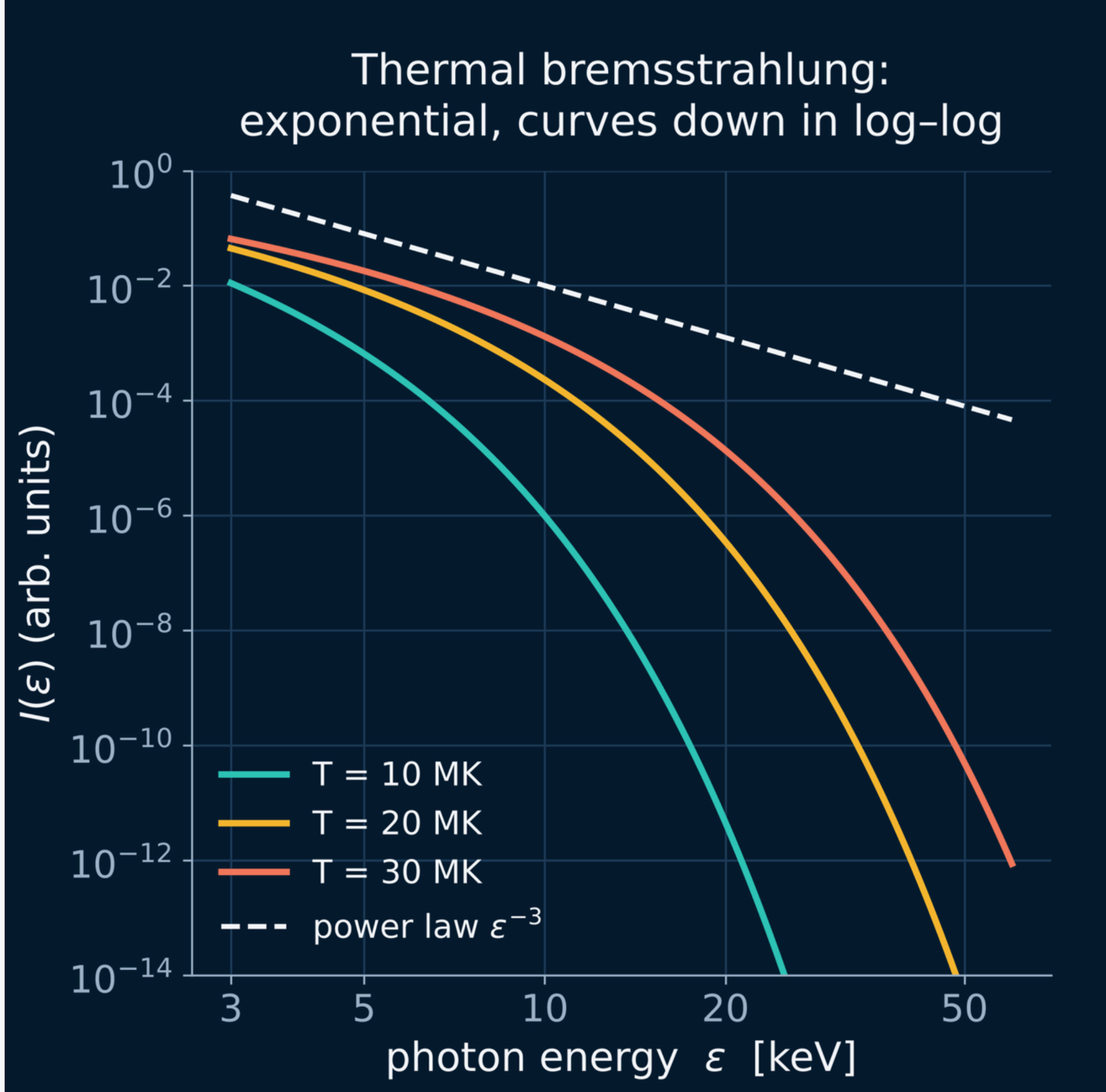


Figure: own work. Illustrative model

Thin vs. thick target



How much energy do electrons lose where they emit?

Thin target

- Electrons lose little energy in the source
- Tenuous coronal sources
- Photon index $\gamma = \delta + 1$

Thick target (Brown 1971)

- Electrons stop completely by collisions
- Dense chromospheric footpoints
- Photon index $\gamma = \delta - 1$

TAKE-AWAY Same electrons, different spectra: the target model sets how we read δ from γ.

$$\gamma_{\mathrm{obs}} \xrightarrow{\text{thick target}} \delta = \gamma + 1$$

$$\xrightarrow{\text{leaky box}} \frac{\tau_a}{\tau_{\mathrm{esc}}} = \delta(\delta - 1)$$

From photons back to acceleration

Measure γ above the thermal component, turn it into the electron index δ with a target model, then into a ratio of time scales with an acceleration model. Every arrow is a model with its own assumptions.

Reference: Brown (1971); Holman et al. (2011), Space Sci. Rev. 159, 107

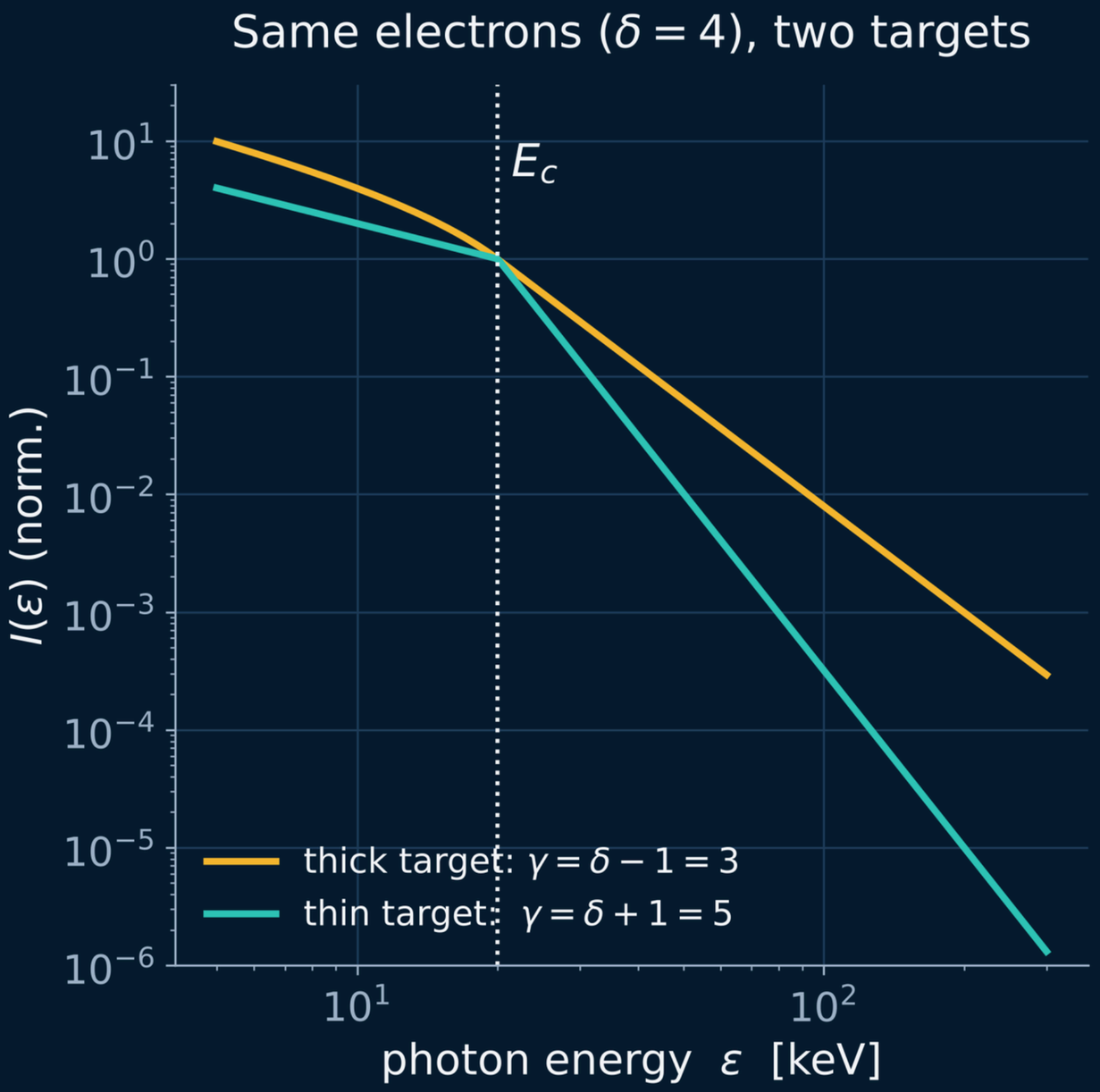


Figure: own work. Illustrative model

Two pictures of a flare



Classic thick target vs. stochastic acceleration

Classic thick target

- Acceleration and emission are separate
- Injected spectrum is an input
- Needs huge electron numbers

Stochastic acceleration

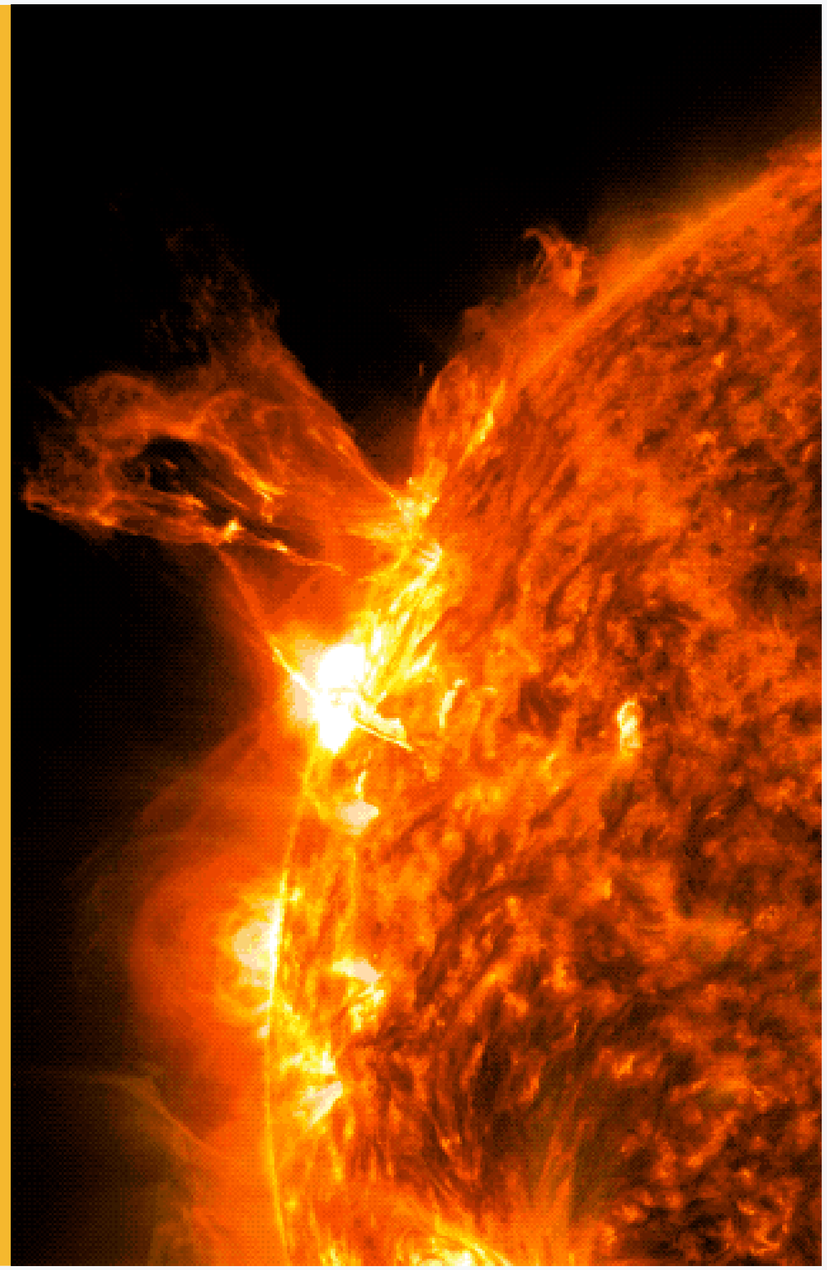
- Acceleration, trapping and escape in one region
- Spectrum is an output: δ set by $\tau a/\tau esc$
- Needs the turbulence Dpp, hard to measure

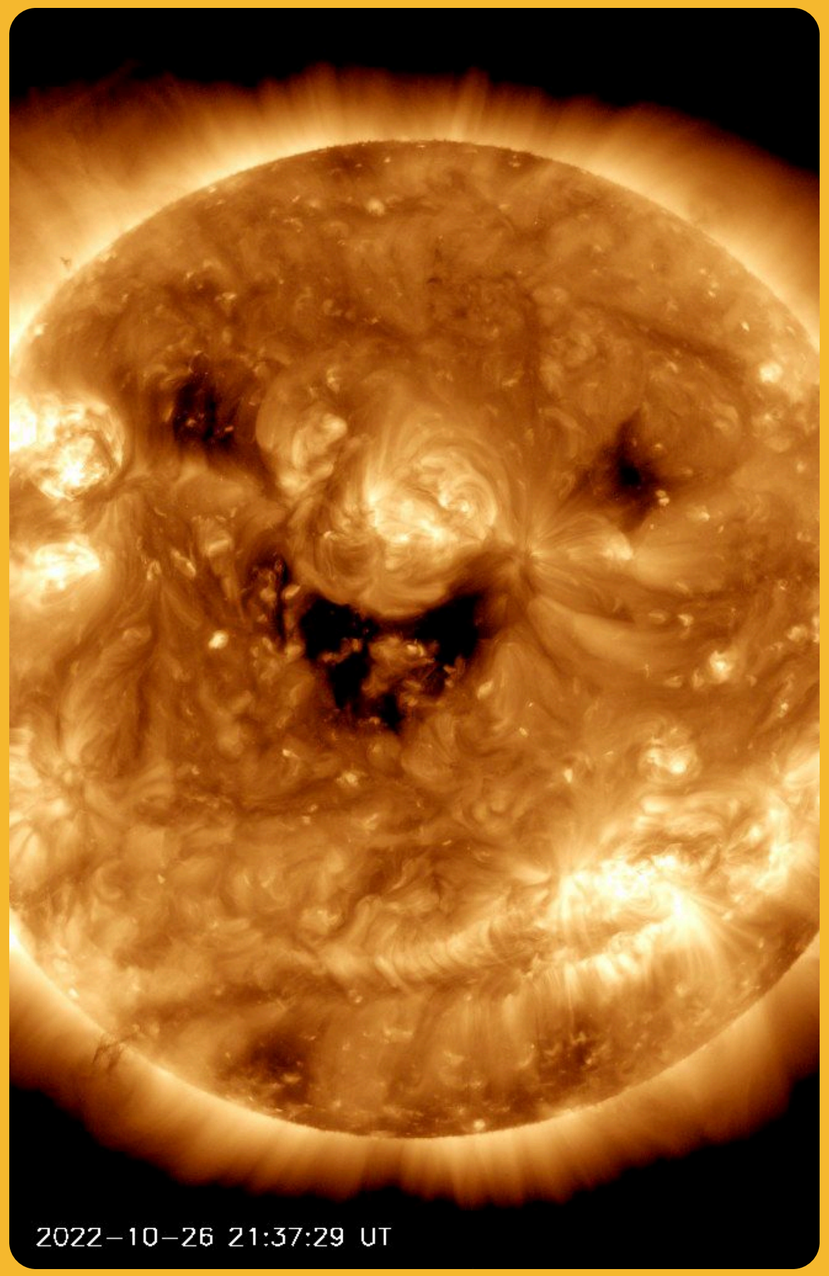
TAKE-AWAY Not exclusive: Fokker-Planck is the common language to write and test both.

Conclusions

Summary

1. Flares make two electron populations: a hot Maxwellian core and a power-law tail.
2. Langevin (one electron) and Fokker–Planck (the population) describe both with drift + diffusion.
3. Spectral indices measure competing time scales, read through bremsstrahlung models.





Thanks!

Questions?

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